

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 8: DNS Spoofing Forensics

Three-Week Delivery Block 3/3 | 19-25 September 2026

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Important Pre-Lab Instruction Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.	
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 8
Lab Title	DNS Spoofing Forensics
Delivery Block	3/3 of 3
Scheduled Dates	19-25 September 2026
Estimated Duration	4-5 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	Controlled DNS-spoof PCAPNG plus baseline DNS/ARP evidence

1. Source Materials and Slide Alignment

- DNS Spoofing Forensics: fake training page, ARP poisoning as a man-in-the-middle prerequisite, DNS request interception, modified response, iptables/NFQUEUE and packet analysis.
- The uploaded slide uses arp.py and dns_spoof.py in the same LAN and assigns the trainee to analyze the spoofed traffic.
- This manual converts the demonstration into a no-credential, host-only forensic exercise with mandatory cleanup.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

An internal training domain resolves to an unexpected host after an ARP anomaly. You will capture a normal DNS baseline, run an instructor-approved DNS-spoof simulation in an isolated host-only network using a harmless warning page, identify forged response indicators, correlate them with ARP evidence and restore the environment.

3. Learning Outcomes

- Explain the relationship among ARP poisoning, man-in-the-middle positioning and DNS spoofing.
- Capture baseline DNS and ARP evidence before a controlled simulation.
- Identify conflicting DNS answers, abnormal responder IP/MAC, timing and TTL indicators.
- Correlate a spoofed DNS response with the web connection that follows.
- Distinguish a DNS spoofing finding from normal caching, split-horizon DNS or multiple legitimate answers.
- Restore forwarding, iptables and ARP state after the lab.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- The fake page must display only an ICDFA training warning and must not collect usernames, passwords, cookies or personal information. Use a reserved training domain such as portal.icdfa.test or bank.training.invalid, never a real bank or public domain.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Virtual machines	Kali analyst plus victim VM	Run on a host-only/internal network.
Web server	Apache2 on analyst VM	Host harmless training page.
Capture tools	Wireshark/tshark	Capture DNS, ARP and HTTP.
Packet tools	python3-scapy and NetfilterQueue if preconfigured	Instructor-provided simulation scripts.
Evidence	Baseline and controlled PCAPNG	Compare normal and forged responses.
Test domain	Instructor-assigned .test/.invalid name	Avoid impersonating real organizations.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab8
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```
sudo apt update
sudo apt install -y apache2 tshark wireshark python3-scapy python3-pip build-essential python3-dev libnetfilter-queue-dev
sudo systemctl enable --now apache2
printf '<!DOCTYPE html>\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: YOUR NAME</p></body></html>\n' | sudo tee /var/www/html/index.html
sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt
ip -br address | tee reports/interfaces.txt
ip route | tee reports/routes.txt
ip neigh show | tee reports/arp_before.txt
```

```

kali@kali: ~/SBT-DF203-Lab8
Session Actions Edit View Help

(kali@kali)~$ mkdir -p ~/SBT-DF203-Lab8/{evidence,working,exported,reports,screenshots,scripts}

(kali@kali)~$ cd ~/SBT-DF203-Lab8

(kali@kali)~/SBT-DF203-Lab8$ pwd
/home/kali/SBT-DF203-Lab8

(kali@kali)~/SBT-DF203-Lab8$ find . -maxdepth 1 -type d -print
.
./exported
./evidence
./working
./screenshots
./scripts
./reports

(kali@kali)~/SBT-DF203-Lab8$ sudo apt install -y apache2 tshark wireshark python3-scapy python3-pip build-essential python3-dev libnetfilter-queue-dev
[sudo] password for kali:
apache2 is already the newest version (2.4.68-1).
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
python3-scapy is already the newest version (2.7.0+dfsg1-1).
python3-scapy set to manually installed.
build-essential is already the newest version (12.12).
build-essential set to manually installed.
python3-dev is already the newest version (3.14.6-1).
python3-dev set to manually installed.
The following packages were automatically installed and are no longer required:
gdal-data libcrypt-dev libhdf4-0 libmbedcrypto16 libproj25 libzxing3 python3.13-minimal
gdal-plugins libfreexl1 libhdf5-310 libmjpegutils-2.1-0t64 libpython3.13-dev onboard-common python3.13-venv
gir1.2-ayatanaappindicator3-0.1 libfyba0t64 libhdf5-hl-310 libmpeg2encpp-2.1-0t64 libraw23t64 proj-bin rpcsvc-proto
gir1.2-girepository-2.0 libgavl1 libjs-jquery-ui libmplex2-2.1-0t64 librttopo1 proj-data samba-ad-provision
libaec0 libgdal37 libjs-underscore libmuparser2v5 libspatialite8t64 python3-fs samba-dsdb-modules
libarmadillo14 libgeos-c1t64 libjxr-tools libnetcdf22 libwireshark18 python3-gpg tdb-tools
libarpac2t64 libgeos3.14.0 libjxr0t64 libodbc2 libwiretap15 python3-talloc unixodbc-common
libavc1394-0 libgeotiff5 libkmlbase1t64 libodbcrc2 libwmflite-0.2-7 python3-tdb
libblosc2-4 libgirepository-1.0-1 libkmlengine1t64 libodbcinst2 libwsutil16 python3.13
libcfitsio10t64 libpgpme45 libkmlengine1t64 libportmidi0 libxerces-c3.2t64 python3.13-dev

libcfitsio10t64 libpgpme45 libkmlengine1t64 libportmidi0 libxerces-c3.2t64 python3.13-dev
Use 'sudo apt autoremove' to remove them.

Upgrading:
dpkg libnetfilter-queue1 libnfnfnetlink0 python3-pip python3-pip-whl

Installing:
libnetfilter-queue-dev

Installing dependencies:
libnfnfnetlink-dev libpkgconf7 pkgconf pkgconf-bin

Suggested packages:
libnetfilter-queue-doc

Summary:
Upgrading: 5, Installing: 5, Removing: 0, Not Upgrading: 2098
Download size: 3,039 kB / 4,572 kB
Space needed: 356 kB / 54.1 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 libnfnfnetlink0 amd64 1.0.2-3+b2 [14.9 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libnetfilter-queue1 amd64 1.0.5-4+b2 [15.0 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 libpkgconf7 amd64 2.5.1-4 [47.8 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 pkgconf-bin amd64 2.5.1-4 [35.9 kB]
Get:5 http://kali.download/kali kali-rolling/main amd64 pkgconf amd64 2.5.1-4 [33.6 kB]
Get:6 http://http.kali.org/kali kali-rolling/main amd64 libnfnfnetlink-dev amd64 1.0.2-3+b2 [8,220 B]
Get:7 http://http.kali.org/kali kali-rolling/main amd64 libnetfilter-queue-dev amd64 1.0.5-4+b2 [18.5 kB]
Get:8 http://http.kali.org/kali kali-rolling/main amd64 python3-pip all 26.1.2+dfsg-1 [1,402 kB]
Get:9 http://http.kali.org/kali kali-rolling/main amd64 python3-pip-whl all 26.1.2+dfsg-1 [1,463 kB]
Fetched 3,039 kB in 5s (669 kB/s)
(Reading database ... 420551 files and directories currently installed.)
Preparing to unpack .../dpkg.1.23.7+kali1_amd64.deb ...
Unpacking dpkg (1.23.7+kali1) over (1.22.21+kali1) ...
Setting up dpkg (1.23.7+kali1)...
(Reading database... 420552 files and directories currently installed.)
Preparing to unpack .../0-libnfnfnetlink0_1.0.2-3+b2_amd64.deb...
Unpacking libnfnfnetlink0:amd64 (1.0.2-3+b2) over (1.0.2-3)...
Preparing to unpack .../1-libnetfilter-queue1_1.0.5-4+b2_amd64.deb...
Unpacking libnetfilter-queue1:amd64 (1.0.5-4+b2) over (1.0.5-4+b1)...
Selecting previously unselected package libpkgconf7:amd64.
Preparing to unpack .../2-libpkgconf7_2.5.1-4_amd64.deb...
Unpacking libpkgconf7:amd64 (2.5.1-4)...
Selecting previously unselected package pkgconf-bin.

```

```

Session Actions Edit View Help
Processing triggers for kali-menu (2025.3.2)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

(kali@kali)-[~/SBT-DF203-Lab8]
$ sudo systemctl enable --now apache2

(kali@kali)-[~/SBT-DF203-Lab8]
$ printf '%!DOCTYPE html%\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: YOUR NAME</p></body></html>\n' | sudo tee /var/www/html/index.html
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: YOUR NAME</p></body></html>

(kali@kali)-[~/SBT-DF203-Lab8]
$ sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt
996ad035a8e146a61eced04db274ec6073309378a41c9e3256d0aab9d2a2df41 /var/www/html/index.html

(kali@kali)-[~/SBT-DF203-Lab8]
$ ip -br address | tee reports/interfaces.txt
lo                UNKNOWN      127.0.0.1/8  ::1/128
eth0              UP          192.168.225.128/24  fe80::92cd:feed:c2d:71f9/64
eth1              UP          192.168.88.130/24  fe80::afe1:5436:8b42:335d/64
docker0           DOWN        172.17.0.1/16

(kali@kali)-[~/SBT-DF203-Lab8]
$ ip route | tee reports/routes.txt
default via 192.168.225.2 dev eth0 proto dhcp src 192.168.225.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.88.0/24 dev eth1 proto kernel scope link src 192.168.88.130 metric 101
192.168.225.0/24 dev eth0 proto kernel scope link src 192.168.225.128 metric 100

(kali@kali)-[~/SBT-DF203-Lab8]

```

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab8-Samuel_Chinedu_Onu
Trainee name	Samuel Chinedu Onu
Date and time started	23 rd Sept 2026, 3pm
Evidence file name(s)	/var/www/html/index.html reports/training_page_sha256.txt reports/interfaces.txt reports/routes.txt reports/arp_before.txt
Source or generation method	Generated on Kali linux VM during lab setup
Original SHA-256	996ad035a8e146a61eced04db274ec6073309378a41c9e3256d0aab9d2a2df41
Working-copy SHA-256	996ad035a8e146a61eced04db274ec6073309378a41c9e3256d0aab9d2a2df41
Analysis workstation/VM	Kali linux VM
Notes on any changes	No PCAP captures generated yet